

Matheus Eduardo Garbelini, Ph.D

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Employment History

- Nov 2025 – Ongoing **Lead Research Scientist. Programme Manager for Wireless Security.** Nanyang Technological University (NTU), Center for Communication and Signal Processing, Temasek Laboratories@NTU.
- Oct 2023 – Nov 2025 **Post-doctoral Research Fellow.** Singapore University of Technology and Design (SUTD), Member of ASSET Research Group.
- Jan 2019 – Sep 2019 **Research Assistant.** Singapore University of Technology and Design (SUTD), Member of ASSET Research Group.
- Dec 2017 – Dec 2018 **R&D Engineering Intern.** Lactec. Curitiba, Brazil.
- Jan 2016 – Dec 2017 **R&D Engineering Intern.** Zulgg Tecnologias. São José dos Pinhais, Brazil.

Education

- Sep 2019 – Sep 2023 **Ph.D., Singapore University of Technology and Design**, Computer Science. Thesis title: *Directed Stateful Fuzzing of Wireless Protocols*. * Received Outstanding Thesis Award.
- Feb 2014 – Dec 2018 **B.E. Electronic Engineering, Pontifical Catholic University of Paraná (PUCPR)**, Curitiba, Brazil. * Received Best Student Award.

Key Research Contributions

- 2024 **IoT Defense.** Proposed, designed and evaluated technology for defending Bluetooth Low Energy (BLE) devices against over-the-air attacks (paper in ACSAC 2024).
- 2019-2025 **IoT/Wireless Software Testing via Fuzzing.** Proposed, designed and evaluated some of the pioneering technologies for Wireless Protocol Fuzzing for 5G NR, Bluetooth BR/EDR, Bluetooth Low Energy (BLE), Zigbee etc (papers in USENIX ATC 2020, USENIX Security 2022, IEEE TDSC 2020, ICST 2024, TDSC, etc.).

Awards and Honours

- 2024 **2024 Outstanding Thesis Award** from Singapore University of Technology and Design for PhD thesis "Directed Stateful Fuzzing of Wireless Protocols".
- 2023 **36,000 USD award** from [Qualcomm Inc.](#) and [MediaTek Inc.](#) for finding a total of 12 security flaws (7 with **High severity**) in 5G baseband modems.
- 2022 **6,000 USD award** from [Intel](#) for discovering two Intel AX200 attacks under the BrakTooth family of security vulnerabilities.
- 2020 **Recognized by Medtronic Inc.** as an outstanding research contributor for discovering Sweyn-Tooth family of vulnerabilities.
- 2018 **Awarded with PUCPR Marcelino Champagnat award** for best student in Electronic Engineering class of 2018 and PUCPR Deans's list of 2017 for remarkable academic performance.

Broader Research Impact

- 2019-2025  Works on Wireless Fuzzing have discovered over [70 security vulnerabilities](#) across Wi-Fi, Bluetooth Low Energy, Bluetooth BR/EDR, CoAP, Zigbee and 5G NR implementations.
- 2020-2024  Research has been featured in [WIRED](#), [HackerNews](#), [ThreatPost](#), [PC Magazine](#), [Security-Week](#) along with **100+ news articles** worldwide. Such research has also raised alerts from regulatory agencies like [CSA \(Singapore\)](#), [HSA \(Singapore\)](#), [IMDA \(Singapore\)](#), [CISA \(Department of Homeland Security\)](#), [FDA \(USA\)](#) etc.
- 2019-2023  Research on Wireless fuzzing (Wi-Fi, Bluetooth BR/EDR and Bluetooth Low Energy) has been translated by Keysight Technologies in their [IoT Security Assessment Software](#).
- 2023  Discovery of 5G security vulnerabilities (5Ghoul) [featured by Channel News Asia](#).
- 2020-2022  [Recognized by Cyber Security Agency \(CSA\) Singapore](#) for discovering SweynTooth and BrakTooth family of security vulnerabilities.

Notable Open-source Contribution

- 2024  [U-Fuzz](#): Over-the-air IoT protocol Fuzzing tool. Experimented to find 11 new vulnerabilities across CoAP, Zigbee and 5G NR implementations. **Github ★: 70+**.
- 2023  [5Ghoul](#): Over-the-air 5G SA exploits and 5G NR fuzzing tool. Independently used by industry (pen testing companies) to find 5G SA vulnerabilities. **Github ★: 500+**.
- 2022  [BrakTooth](#): Over-the-air Bluetooth BR/EDR exploits. Independently used by industry (e.g., Samsung and MediaTek) to find Bluetooth BR/EDR vulnerabilities. **Github ★: 430+**.
- 2022  [Bluetooth BR/EDR Sniffer](#): Reverse engineered Bluetooth BR/EDR sniffer. **Github ★: 490+**.
- 2020  [SweynTooth](#): A set of Bluetooth Low Energy (BLE) exploits. Independently used by industry (e.g., EM Semiconductor, Microchip) and government agencies (e.g., regulatory agencies) to find BLE security vulnerabilities. **Github ★: 260+**.

Selected Key Publications

1. *Sni5Gect: A Practical Approach to Inject aNRchy into 5G NR*. Shijie Luo and [Matheus Garbelini](#) and Sudipta Chattopadhyay and Jianying Zhou 34th USENIX Security Symposium (USENIX Security 25).
2. *5Ghoul: Unleashing Chaos on 5 G Edge Devices Via Stateful Multi-Layer Fuzzing*. [Matheus E. Garbelini](#), Zewen Shang, Shijie Luo, Sudipta Chattopadhyay, Sumei Sun and Ernest Kurniawan. IEEE Transactions on Dependable and Secure Computing (TDSC), 2025.
3. *BrakTooth: Causing Havoc on Bluetooth Link Manager via Directed Fuzzing*. [Matheus E. Garbelini](#), Vaibhav Bedi, Sudipta Chattopadhyay, Sumei Sun, and Ernest Kurniawan. USENIX Security Symposium 2022.
4. *SweynTooth: Unleashing Mayhem over Bluetooth Low Energy*. [Matheus E. Garbelini](#), Chundong Wang, Sudipta Chattopadhyay, Sumei Sun, and Ernest Kurniawan. USENIX Annual Technical Conference (USENIX ATC) 2020.

Research Publications

Google Scholar: https://scholar.google.com/citations?user=_PXBxu0AAAAJ&hl=en

Maintained List: <https://matheus-garbelini.github.io/home/publication/>

Patents

- 2022  [Method and System for Detecting Anomalies of Server and Client](#), US Patent no. US11349963B1. Inventor: Sudipta Chattopadhyay (SUTD), **Matheus Eduardo Garbelini (SUTD)**, Francis-Bs Ngian (Keysight Technologies) and Cyril Tan (Keysight Technologies).

Research Grants I Helped to Write

- 2023-2026  **MOE Tier 2 Award**, Towards Automated Fuzzing and Debugging of Closed Communication Protocol Stacks (550K SGD).
- 2022-2025  **Future Communications Program (FCP)**, Towards Differential Fuzzing of 5G and beyond 5G Communications (967K SGD).
- 2020-2023  **NRF National Satellite of Excellence in Trustworthy Systems and Software**, Over-the-air Security Testing of Wireless Protocol Implementations (548K SGD).
- 2021-2023  **NRF National Satellite of Excellence in Trustworthy Systems and Software**, Towards a Universal IoT Protocol Testing Infrastructure (398K SGD).

Invited Presentations, Talks & Sessions

- Oct 2024  [Panel: Secured Communication Networks](#), World Telecommunication Standardization Assembly, New Delhi.
- August 2024  [5Ghoul Framework - 5G NR Attacks & 5G OTA Fuzzing](#), DEFCON32, Las Vegas.
- May 2024  [IoT and 5G Security: Discovering Wireless Protocol Vulnerability through Fuzzing](#), Singapore ISC² Chapter.
- Jan 2024  [Small group QA Session Facilitator](#), Singapore Global Young Scientists Summit (GYSS).
- October 2020  [Invited Tutorial on IoT Security](#), Singapore International Cyber Week, 2020.

Teaching Assistance

- 2024  **Software Testing and Verification**, Class size = 35
- 2020  **Systems Security**, Class size = 45

Advised Students (As a Postdoc.)

Current PhD Students

- Sept 2023-Present  **Shijie Luo**. *A Practical Approach to Inject Attacks into 5G NR*. **Ongoing work**. First Author paper in **34th USENIX Security Symposium (USENIX) 2025**.
-  **Guoqiang Hua**. *Automated Exploit Generation and Debugging for Over-the-air Attacks*. First Author paper in **Annual Computer Security Applications Conference (ACSAC) 2024**.
-  **Geovani Benita**. *Efficient Link-layer Attacks and Defence for Wireless Network*. First Author paper in **Proceedings of the 19th ACM Conference on Security and Privacy in Wireless and Mobile Networks (WiSec) 2026**.

Professional Service

Journal Review

- 2025  [ACM Computing Surveys](#)

Professional Service (continued)

Technical Program Committee

-  [DATE 2025, ACM CPSS 2025.](#)
- 2024  [ICCD 2024.](#)
- 2023  [SOLI 2023.](#)